

Supermarket Sales Analysis & Business Insights

Data Analytics with AI Internship Program 2026

Project Topic: Supermarket Sales Analysis & Business Insights

Domain: Data Analytics with AI

1. Introduction

This project demonstrates a Python-based data analytics workflow for supermarket transaction data. The analysis studies sales performance, product lines, branches, payment methods and customer patterns.

2. Objectives

- Inspect and clean transaction data.
- Calculate KPIs.
- Compare product-line and branch sales.
- Analyze payment and gender patterns.
- Create visualizations and business insights.

3. Tools and Technologies

Python, Pandas, NumPy, Matplotlib and Seaborn.

4. Methodology

The workflow consists of data loading, quality inspection, cleaning, aggregation, visualization and insight generation. Grouped summaries are produced for product line, branch/city, payment method and gender.

5. Key Results

Metric	Result
Total sales	1265.50
Average transaction	63.27
Average rating	7.39
Top product line	Food and beverages
Top branch	C - Naypyitaw
Top payment method	Credit card

6. Business Insights

The highest-sales product line in the included dataset is **Food and beverages**. The highest-sales branch grouping is **C - Naypyitaw**, and the leading payment method by sales is **Credit card**. These comparisons can support inventory planning, branch-level review and payment-channel decisions.

7. Conclusion

The project shows how transaction data can be converted into KPIs, visual comparisons and concise business insights. The same workflow can be applied to a larger official internship dataset.

8. Deliverables

Source code, dataset, requirements.txt, generated analytical outputs, README.md and this project report.